

LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 3
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1. TITLE

Student Profile Manager Using JavaScript Class and DOM Manipulation

2. AIM

To build an interactive Student Profile Manager application using JavaScript ES6 classes, constructor methods, input fields, and DOM dynamic HTML creation.

3. OBJECTIVE

The objectives of this experiment are:

- To define ES6 Student class with constructor properties.
- To capture user input from text and number input fields.
- To dynamically instantiate objects and update DOM innerHTML.
- To style modern UI cards with border shadow and centered alignment.

4. THEORY

JavaScript ES6 classes provide object-oriented structure. DOM manipulation using `document.getElementById` and `innerHTML` enables dynamic content generation without full page reloads.

Application Flow:

User Input -> ES6 Student Class Instantiation -> Object Properties -> DOM innerHTML Injection -> Profile Card Output

5. ALGORITHM / PROCEDURE

1. Create task3.html with input fields for Name, Roll No, Department, and CGPA.
2. Define Student class with constructor(name, roll, department, cgpa).
3. Implement createProfile() function to read input values.
4. Instantiate new Student object with input parameters.
5. Format and inject profile HTML into the #profile container div.

6. PROGRAM

task3.html

```
<!DOCTYPE html>
<html>

<head>

  <title>Student Profile Manager</title>

  <style>

    body {
      font-family: Arial;
      background-color: #eaf4ff;
      text-align: center;
      margin: 40px;
    }

    h1 {
      color: #174a7e;
    }

    input {
      width: 380px;
      padding: 10px;
      margin: 6px;
      border: 1px solid #aaa;
      border-radius: 5px;
    }

    button {
      background-color: #174a7e;
      color: white;
      padding: 10px 25px;
      border: none;
      border-radius: 5px;
      margin: 8px;
    }

    #profile {
      width: 430px;
      margin: 25px auto;
      padding: 20px;
      background-color: white;
      border: 2px solid #174a7e;
      border-radius: 8px;
      box-shadow: 3px 3px 8px #aaa;
      text-align: left;
    }

    #profile h2 {
      color: #174a7e;
      text-align: center;
    }

    #profile p {
      font-size: 16px;
    }

  </style>

</head>

<body>

  <h1>Student Profile Manager</h1>

  <input type="text" value="Name" />
  <input type="text" value="Roll No." />
  <input type="text" value="Email" />
  <input type="text" value="Phone" />
  <input type="text" value="Address" />
  <input type="text" value="City" />
  <input type="text" value="State" />
  <input type="text" value="Zip" />
  <input type="text" value="DOB" />
  <input type="text" value="Gender" />
  <input type="text" value="Blood Group" />
  <input type="text" value="Hobbies" />
  <input type="text" value="Comments" />
  <input type="button" value="Save" />
  <input type="button" value="Cancel" />

  <div id="profile">
    <h2>Student Profile</h2>
    <p>Name: <input type="text" value="Name" /></p>
    <p>Roll No.: <input type="text" value="Roll No." /></p>
    <p>Email: <input type="text" value="Email" /></p>
    <p>Phone: <input type="text" value="Phone" /></p>
    <p>Address: <input type="text" value="Address" /></p>
    <p>City: <input type="text" value="City" /></p>
    <p>State: <input type="text" value="State" /></p>
    <p>Zip: <input type="text" value="Zip" /></p>
    <p>DOB: <input type="text" value="DOB" /></p>
    <p>Gender: <input type="text" value="Gender" /></p>
    <p>Blood Group: <input type="text" value="Blood Group" /></p>
    <p>Hobbies: <input type="text" value="Hobbies" /></p>
    <p>Comments: <input type="text" value="Comments" /></p>
  </div>

</body>

</html>
```

```
</style>

</head>

<body>

  <h1>Student Profile Manager</h1>

  <h2>Enter Student Details</h2>

  <input type="text" id="name" placeholder="Enter Name">
  <br>

  <input type="text" id="roll" placeholder="Enter Roll Number">
  <br>

  <input type="text" id="department" placeholder="Enter Department">
  <br>

  <input type="number" id="cgpa" placeholder="Enter CGPA">
  <br>

  <button onclick="createProfile()">Create Profile</button>

  <div id="profile"></div>

  <script>

    class Student {

      constructor(name, roll, department, cgpa) {
        this.name = name;
        this.roll = roll;
        this.department = department;
        this.cgpa = cgpa;
      }

    }

    function createProfile() {

      var name = document.getElementById("name").value;
      var roll = document.getElementById("roll").value;
      var department = document.getElementById("department").value;
      var cgpa = document.getElementById("cgpa").value;

      var student = new Student(name, roll, department, cgpa);

      var profile = document.getElementById("profile");

      profile.innerHTML =
        "<h2>Student Profile</h2>" +
        "<p><b>Name</b> : " + student.name + "</p>" +
        "<p><b>Roll No</b> : " + student.roll + "</p>" +
        "<p><b>Department</b> : " + student.department + "</p>" +
        "<p><b>CGPA</b> : " + student.cgpa + "</p>";

    }

  </script>
```

```
</body>

</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
class Student { constructor(...) }	Defines an ES6 class blueprint with constructor method.
document.getElementById("name").value	Extracts current value entered in the text input field.
new Student(name, roll, ...)	Creates a new Student object instance.
profile.innerHTML = "..."	Dynamically populates container element with HTML markup.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Form Input	Enter student details	Input fields accept text and number data	Pass
TC-02	Click Create Profile	Click button	Student object instantiated and profile rendered	Pass
TC-03	DOM Update	Inspect #profile div	Name, Roll No, Department & CGPA display properly	Pass

9. OUTPUT

Student Profile Manager

Enter Student Details

Create Profile

Student Profile

Name : Dinesh Kumar
Roll No : A24126552101
Department : CSM
CGPA : 9.1

Output 1: Student Profile Manager Using JavaScript Class and DOM Manipulation Rendered Webpage

10. RESULT

Thus, student profile manager using javascript class and dom manipulation was successfully implemented and verified across all test cases.